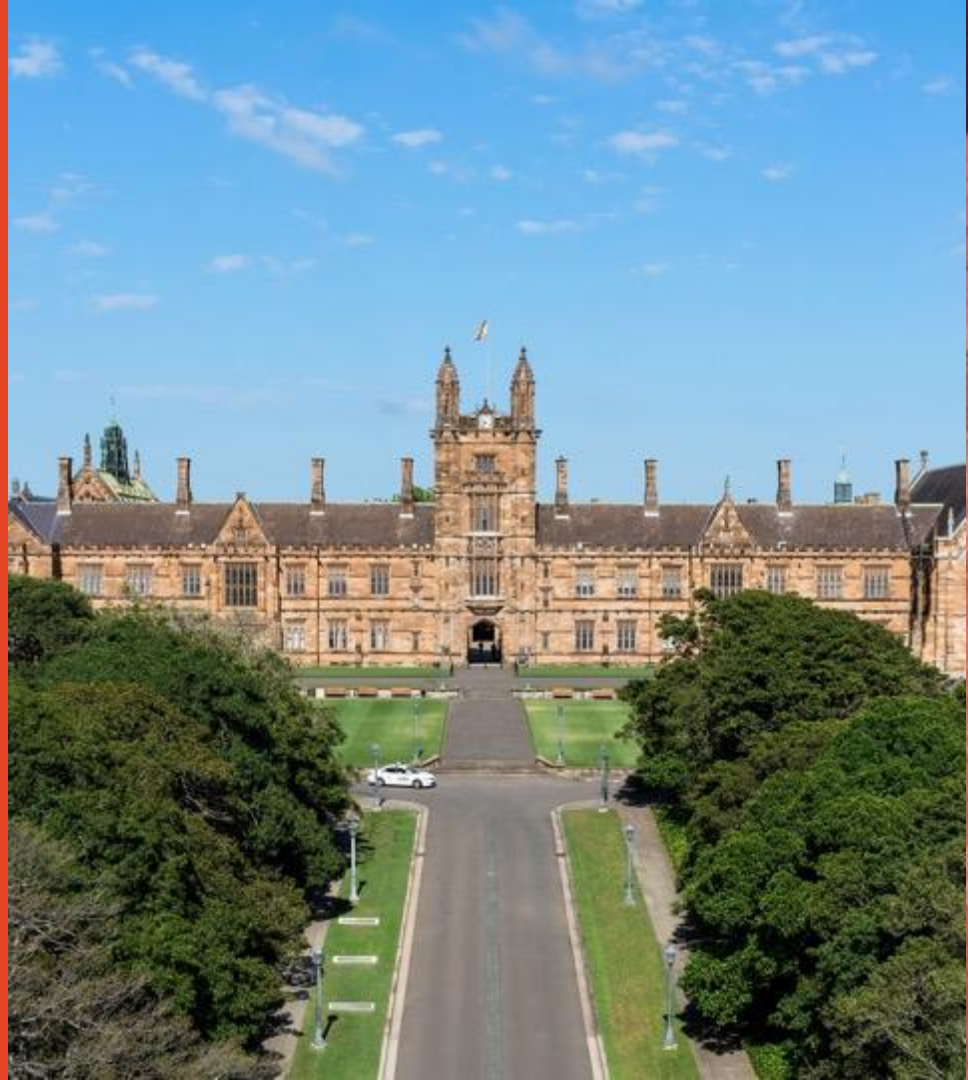
Automated Detection of Pollinator–Flower Interactions from Citizen Science Images

Presented by Group CS46 - 1
Faculty of Engineering,
5703 Group Based Capstone



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Motivation & Research Objective

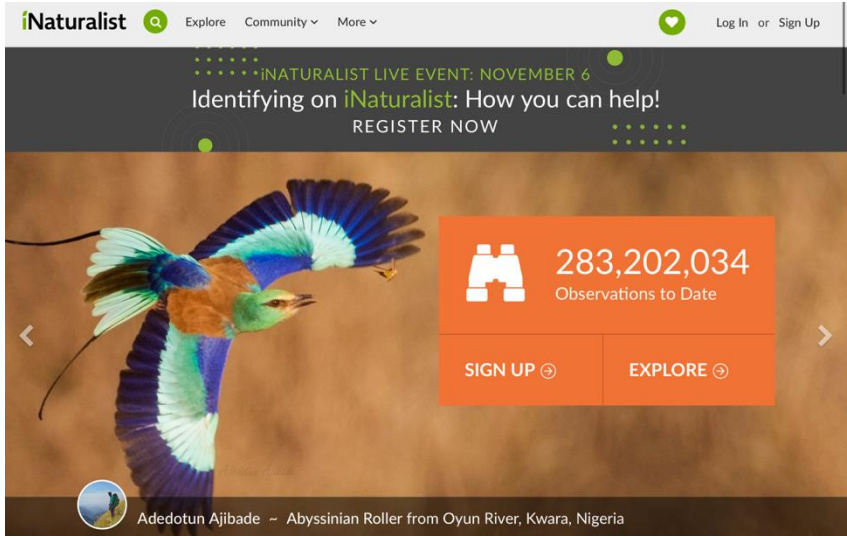
Why This Research Is Needed ?

The Challenge

- Manual annotation of pollinator-flower interactions is labor-intensive and time-consuming, limiting ecological research scale

The Application Gap

- Limited automated tools exist to analyze pollinator preferences and ecological patterns across regions and species



Our Objectives:

- Automate pollinator–flower detection
- Detect floral traits interacted with by pollinators
- Provide a user-friendly web tool for researchers

Background & Related Work

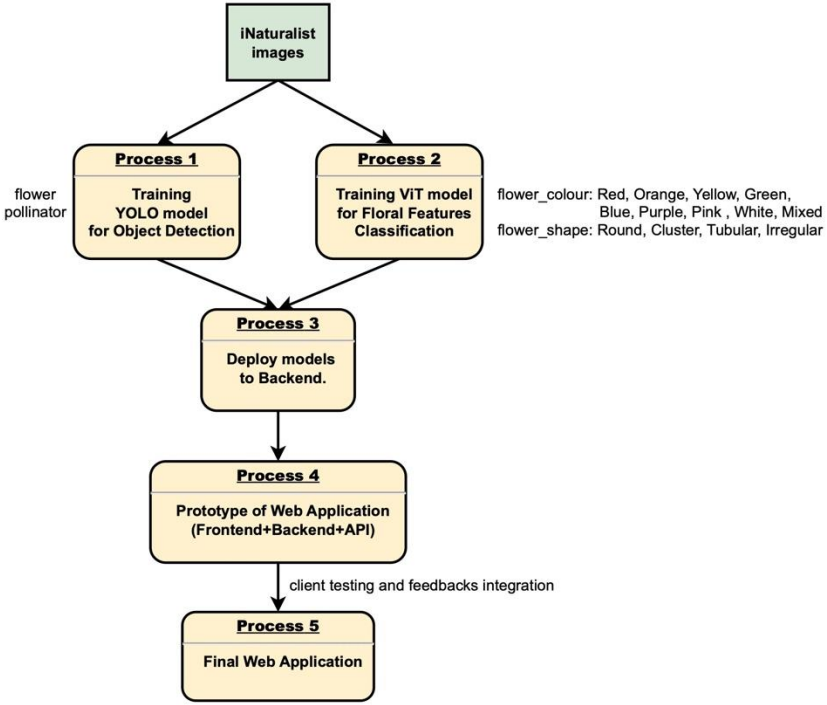
Background

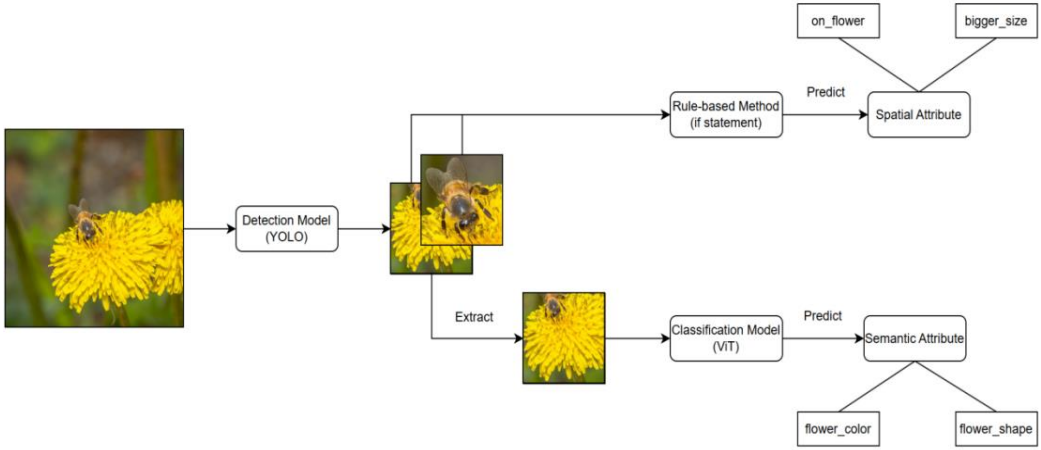
- Pollinators are vital for biodiversity and agriculture. Over 75% of flowering plants depend on them for reproduction.
- Global decline in pollinators urges scalable monitoring tools
- Platforms like iNaturalist provide large image datasets of pollinators and flowers, but manual labeling remains a major bottleneck.

Related Work

- **YOLO:** Recent versions (v8/v10) enable precise, small-object recognition.
- **Vision Transformer:** self-attention for fine-grained classification; excels at identifying detailed floral features.
- **Web Integration:** Previous works emphasize lab models; this project extends them to a web-deployed, user-friendly ecological tool.

Project Workflow



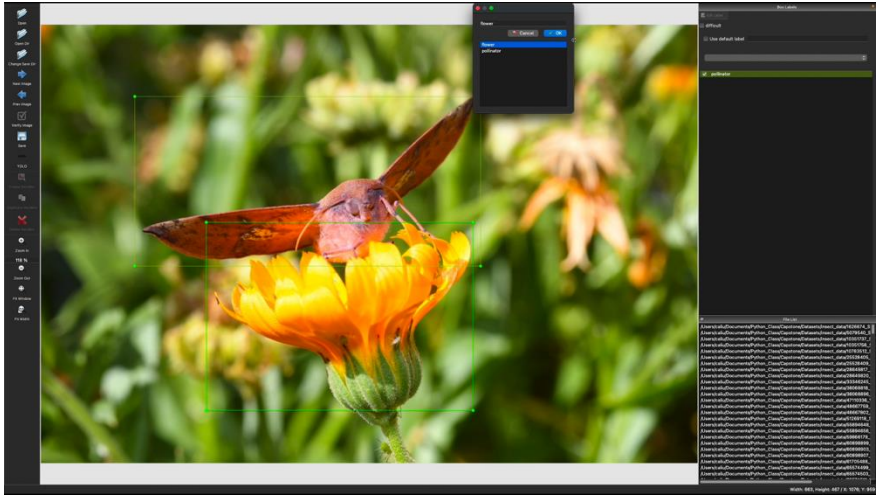


Key Techniques

Dataset Collection & Processing

We gathered a large set of iNaturalist images showing pollinators interacting with flowers. These images come from different environments and lighting conditions to make sure our system can handle various situations.

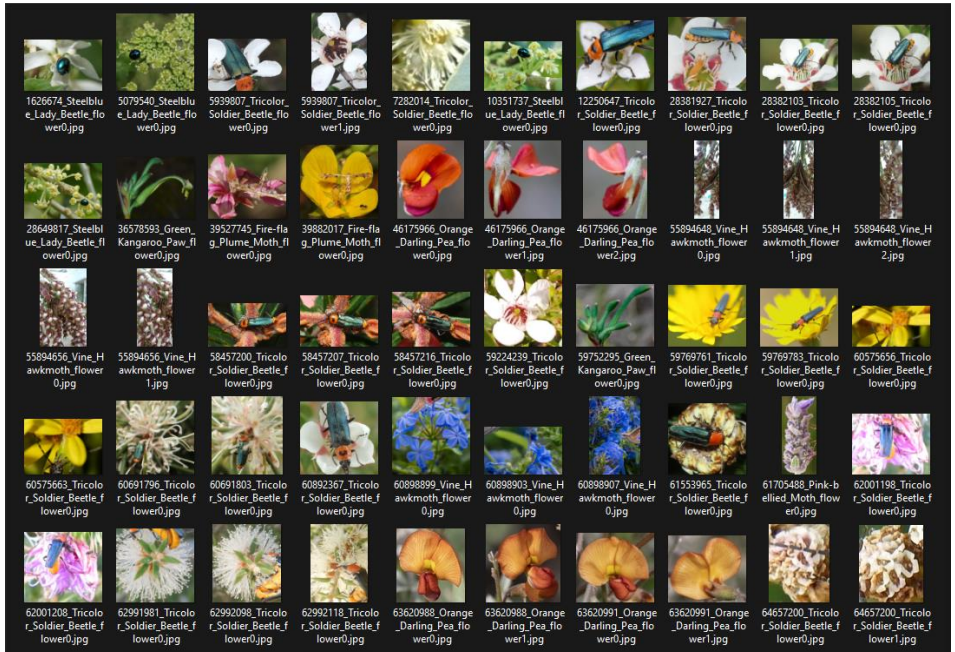
- Training and testing data collected by using a custom web crawler.
- Uitilised external annotation tool to label the images with YOLO bounding boxes for training.

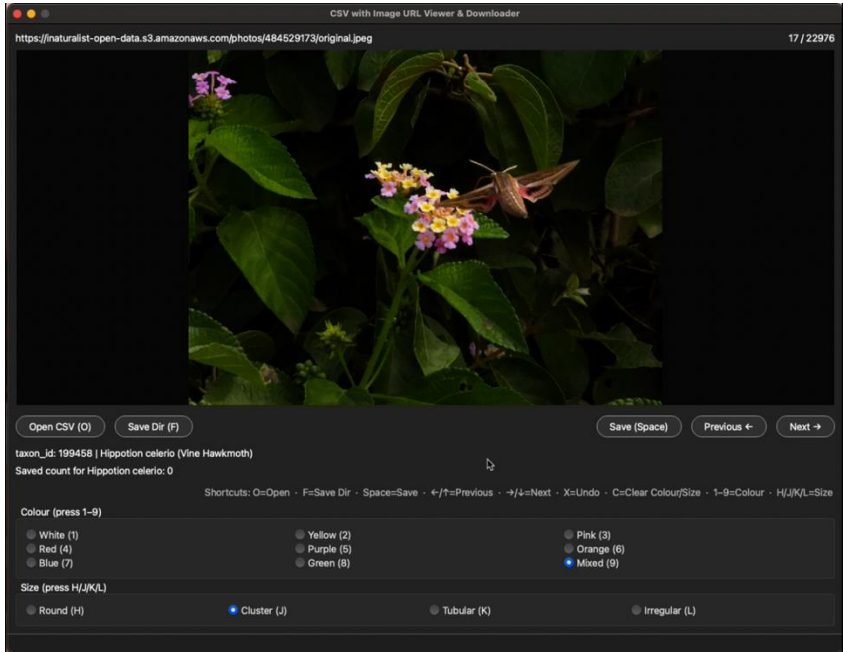


Key Techniques

Dataset Collection & Processing

- Designed a labeling application to assist us to build a reliable datasets.
- Deployed a trained YOLO model to automatically extract flower regions before classification.





Key Techniques

Model Training

- We used these images to teach our AI models how to recognise different objects and recognise different features of flower.
- Think of this as “training” the system to see and understand patterns, just like how people learn to recognise faces or animals over time.

Detection & Classification

- Once trained, the model can automatically detect whether a flower or a pollinator appears in a new image and classify what type it is.
- For example, it can tell if there have any a pollinator around the flower, and the floral features.

Key Techniques

Result Integration

- Finally, the system combines all these detection results into a csv file and visualise on website.
- This allows researchers to see interaction trends — for example, which flowers attract more pollinators or when activity is highest.

System Implementation

Backend

Developed with FastAPI, enabling asynchronous API endpoints for model inference and result retrieval.

Frontend

Built with React, providing an interactive interface for CSV upload, result visualization, and result CSV download.

API Integration

Implemented a RESTful architecture connecting the YOLO and ViT models.

Deployment

Containerised by using Docker, ensuring reproducibility and cross-platform compatibility.

System Key Features

High-throughput Crawler

- Multi-threaded fetching with **thread-safe counter**
- **Thread-pool** controlled concurrency for downloads
- Idempotent steps to avoid duplication under retries

Resilient React Apps (State & Recovery)

- **React Context** as the single source of truth; route changes don't drop data
- Persist **pipelineJobId**; on reload, poll by ID and **rebuild UI** from results
- Robust replay: disciplined timer management, **retry with backoff**, and **batched box fetching** to restore full state

Results & Evaluation

Machine Leaning Model: Over 90% accuracy in predictions

Frontend: User-friendly with rich interactive features

Backend: Only 0.4-second response time per image for analysis

Model Performance

```
=== Testsets Results ===
[Test_Mixed] N=309
  Color Acc:  0.9288 | AUC: 0.9966
  Shape Acc:  0.8738 | AUC: 0.9779
  Strict Acc: 0.8285
  Avg Time / sample: 2.907 ms
[Test_PF_Hard] N=101
  Color Acc:  0.9010 | AUC: 0.9861
  Shape Acc:  0.7525 | AUC: 0.9023
  Strict Acc: 0.6832
  Avg Time / sample: 2.541 ms
[Test_FlowerClean] N=230
  Color Acc:  0.9696 | AUC: 0.9987
  Shape Acc:  0.9217 | AUC: 0.9930
  Strict Acc: 0.9043
  Avg Time / sample: 2.489 ms

=== Overall (weighted over all samples) ===
  N=640
  Color Acc:  0.9391 | AUC: 0.9966
  Shape Acc:  0.8719 | AUC: 0.9785
  Strict Acc: 0.8328
  Avg Time / sample: 2.504 ms

=== Average (macro over 3 testsets) ===
  Color Acc:  0.9331 | AUC: 0.9938
  Shape Acc:  0.8493 | AUC: 0.9578
  Strict Acc: 0.8053
  Avg Time / sample: 2.646 ms
```

Frontend User Interface

Backend Performance Comparison

Version	Execution	Batch	Parallel DL	Delay	Conn Pool	Keepalive	Timeout	Total Time	Avg Time	Speed up
Latest	Async	30	√	0ms	150	75	8	**15.95s**	**0.44s**	**5.91x**
Mid-stage	Async	5	√	0ms	Default	Default	10	38.04s	1.06s	2.45x
Baseline	Sequentia	1	✗	50ms	Default	Default	10	93.57s	2.60s	1x

Discussion & Analysis

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Key insights

Validated hybrid model (YOLO + ViT) with 90% mAP@50 and 93% precision. Successfully processed 22k+ images, proving high reliability and addressing the core research bottleneck.

Performance factors

Driver: Hybrid architecture and ViT fine-tuning.
Obstacles: Poor data quality of iNaturalist (issues such as no flowers in the picture) and imbalanced category distribution (difference in the occurrence frequency of species).

Key limiting factors

Interaction and proximity: This indicator reflects the spatial co-occurrence between pollinator and flowers rather than verified pollination behavior. Its limitation lies in the poor data quality and the lack of ecological context.

Conclusion & Future Work

Project Success & Research Outcomes

1. **High-Performance Detection System:** Achieved 93% precision, 82% recall, and 90% mAP@50 using YOLOv8 and ViT-B/16 for flower–pollinator interaction detection.

2. **End-to-End Automated Workflow:** Integrated detection (YOLOv8) and classification (ViT-B/16) into a web-based application for non-technical ecological researchers.

3. **Robust Model Optimization:** generalization using dropout, label smoothing, and LoRA fine-tuning to overcome overfitting and class imbalance.

4. **Validated and Scalable Research Framework:** Processed 22,000+ images from iNaturalist; achieved reproducible, scientifically reliable, and scalable results for future research.

Conclusion & Future Work

Next Steps & Research Expansion

1. **Dataset Growth and Quality Improvement:** Expand datasets across regions and seasons; integrate metadata (temperature, habitat) for deeper ecological insights.

2. **Advanced Model Development:** Explore hybrid CNN–Transformer architectures and self-supervised learning to reduce manual labeling and improve feature extraction.

3. **Robust Model Optimization:** generalization using dropout, label smoothing, and LoRA fine-tuning to overcome overfitting and class imbalance.

4. **Validated and Scalable Research Framework:** Processed 22,000+ images from iNaturalist; achieved reproducible, scientifically reliable, and scalable results for future research.

Thank you!

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